IBAD ULLAH

+92 335 1651255 | ibadcui@gmail.com | LinkedIn | GitHub | Google Scholar

OBJECTIVE

Researcher in Software Engineering with a strong focus on Artificial Intelligence, Machine Learning, and Deep Learning for medical imaging, smart agriculture, and text analytics. I aim to contribute to impactful research and teaching through developing robust AI models, publishing in reputable venues, and mentoring undergraduate projects in modern software engineering and intelligent systems.

EXPERIENCE

- **Riphah International University, Malakand Campus** *Mar 2025 – Sep 2026*
  Research Assistant Khyber Pakhtunkhwa, Pakistan
  - Conducted research in deep learning, medical imaging, and AI-driven software engineering.
  - Developed and evaluated machine learning models and experimental pipelines for research publications.
  - Contributed to manuscript preparation, literature review, and paper revisions for peer-reviewed venues.
  - Supported research-based learning and mentoring of student projects within the department.

- **Fiverr (Remote)** *Sep 2022 – Present*
  Flutter Developer (Freelance) Remote
  - Developed Flutter applications for Android/iOS with responsive UI and reusable components.
  - Collaborated with clients to deliver prototypes and production-ready features.
  - Improved app quality through testing, optimization, and iterative refinements.

- **EncoderBytes** *Mar 2022 – Aug 2022*
  IT Intern Peshawar IT Park, Pakistan
  - Automated office processes using MS Office tools, reducing manual effort by 20%.
  - Prepared documentation and presentations in Word, PowerPoint, and Excel for project teams.

EDUCATION

- **Riphah International University, Islamabad** *2025 – 2026*
  Post Graduate Diploma in Professional Ethics & Teaching Methodology Islamabad, Pakistan
  - CGPA: 3.65

- **COMSATS University Islamabad** *2025*
  MS Software Engineering Islamabad, Pakistan
  - GPA: 3.46 / 4.00

- **CECOS University of IT and Emerging Sciences** *Sep 2022*
  BS Software Engineering Peshawar, KPK, Pakistan
  - GPA: 3.22 / 4.00

- **Abbottabad International Model School and College** *Aug 2018*
  FSc Pre-Engineering Batkhela, Malakand, KPK

- **FG Public High School Batkhela** *Jun 2016*
  Matric (Science) Batkhela, Malakand, KPK

Publications

1. Shah Faisal, **Ibad Ullah**, Piniel Alphayo Kambey, Abdul Malik, Muhammad Adeel Ejaz, Sajjad Ali Shah, & Yin-Xiong Li. (2026). **Nurr1 deficiency orchestrates a coupled liver–gut pathological axis revealed by multi-omics and deep-learning histopathology**. *Frontiers in Immunology*, 17, 1861669. Published: 16 June 2026. DOI: 10.3389/fimmu.2026.1861669.

2. Imad Ullah, **Ibad Ullah**, & Naseer Ullah. (2026). **Federated Learning Framework for Privacy-Preserving Threat Detection in Distributed IT Systems**. *Majestic International Journal of AI Innovations*, 1, 1–16. Published: 26 May 2026. DOI: 10.65080/mjai.v1.CM2601105004.

3. Shah Faisal, **Ibad Ullah** (Equal Contribution), Piniel Alphayo Kambey, Abdul Malik, & Muhammad Shakeel. (2026). **Revolutionizing hepatic fibrosis staging: A machine learning approach combining clinical, biochemical, and microbiome insights**. *Computers in Biology and Medicine*, 206, 111584. Elsevier. Available online: 9 March 2026. DOI: 10.1016/j.compbiomed.2026.111584.

4. **Ibad Ullah**, Nasir Rashid, Sheraz Babar, & Naeem Iqbal. (2026). **Exploring the Relationship Between Trait Emotional Intelligence and Performance in the Context of SME Software Engineering**. *Journal of Software: Evolution and Process*, 38(1), e70076. First published: 26 Dec 2025. DOI: 10.1002/smr.70076.

5. Shakir Ullah, Imad Ullah, **Ibad Ullah** (Corresponding Author), Naseer Ullah, & Muhammad Taufiq. (2025). **A Novel Modified Relative Discrimination Criterion for Feature Ranking in Text Classification**. *Spectrum of Engineering Sciences*, 3(10), 349–367. Published: 11 October 2025. DOI: 10.5281/zenodo.17318910.

6. Tariq Rahim, **Ibad Ullah**, Aimal Nazir, Muhammad Salih Tanveer, & Muhammad Rohan Qureshi. (2025). **A Deep Learning Approach to PCOS Diagnosis: Two-Stream CNN with Transformer Attention Mechanism**. *Spectrum of Engineering Sciences*, 3(7), 1–20. Published: 2 July 2025. DOI: 10.5281/zenodo.15790016.

7. Imad Ullah, **Ibad Ullah** (Corresponding Author), Khan, S., Mustafa, S., & Aziz, Q. (2024). **Hybrid Machine Learning Model for Early Detection of Cucumber Leaf Curl Disease in Smart Agriculture**. *Siazga Research Journal*, 3(4), 44–55. Published: 31 December 2024. DOI: 10.5281/zenodo.17440338.

Conference Publications

1. **Ibad Ullah**, Muhammad Usman Yaseen, Noor UL Amin, Muhammad Roohan Qureshi, & Syed Ibrahim. (2025). **Explainable Emotion Recognition from Heart Rate Data Using Deep Learning and XGBoost**. *Proceedings of the 27th International Multitopic Conference (INMIC), IEEE*. Published in IEEE Xplore. Link.

2. **Ibad Ullah**,(2026). **Deep Parallel Hybrid Models and Stacking Ensembles for Clinical Diabetes Mellitus Prediction**. Presented at the 2nd International Conference on Artificial Intelligence (2AI-2026), Central University of Kashmir. Springer proceedings (not yet published online).

Academic & Professional Service

- **Reviewer**, International Journal of Electrical and Computer Engineering (IJECE) *Deep Learning & Machine Learning*
- **Reviewer**, 23rd International Conference on Frontiers of Information Technology (FIT 2026)
- **Reviewer**, 2026 International Conference on Applied Artificial Intelligence (2AI-2026), Central University of Kashmir
- **Reviewer**, Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization (Journal)

Academic Projects

- **Sperm Morphology Classification (Xception + CBAM) on SMIDS Dataset** *Jul 2025*
  Deep learning pipeline for sperm morphology classification using attention mechanisms
  - Combined Xception CNN with CBAM attention to improve focus on diagnostically relevant regions.
  - Trained and evaluated on 3,000 high-resolution microscopic images.
  - Achieved 96.2% accuracy and 0.99 AUC with strong macro precision/recall/F1.

- **Brain Tumor Classification (Federated Learning + Synthetic Data)** *Feb 2025*
  Privacy-preserving medical imaging classification with distributed training
  - Built a Federated Learning workflow to train CNN models without sharing raw patient data.
  - Used Stable Diffusion to generate synthetic MRI images to enrich training data.

- **Road Path Hole Detection** *Jun 2024*
  Computer vision-based pothole detection for infrastructure monitoring
  - Trained an object detection model using a Roboflow dataset and evaluated performance for field deployment.

- **Egg Detection (Agriculture Automation)** *May 2024*
  Image processing-based egg detection to reduce manual labor in poultry farms
  - Designed a detection pipeline for automated egg identification to improve collection efficiency.

- **Driver Behavior Detection (Road Safety Project)** *Apr 2023*
  Vision-based risky behavior detection to enhance road safety
  ◦ Developed a detection approach to identify risky driver behaviors and support alerting mechanisms.
- **Augmented Reality Furniture Shop** *Jan 2022*
  ARCore/ARKit-based mobile app for virtual furniture placement
  ◦ Built an AR shopping experience allowing users to visualize furniture in real environments.
- **Object Detection App (Flutter + TensorFlow Lite)** *May 2022*
  Real-time object detection with optimized mobile inference
  ◦ Implemented models including YOLO/SSD/MobileNet and applied quantization for faster inference.
- **Sentiment Analysis Dashboard (Python + NLP)** *Jun 2022*
  Text preprocessing and visualization for sentiment/emotion analysis
  ◦ Built an interactive dashboard using NLTK preprocessing and Matplotlib-based visualization.

Skills

- **AI/ML** Machine Learning, Deep Learning, Model Evaluation, Metrics
- **Computer Vision** Medical Imaging, Classification, Detection Pipelines
- **NLP** Text Classification, Feature Selection, Preprocessing
- **Programming** Python, Flutter/Dart
- **Data** Pandas, Matplotlib, Data Visualization, Data Preprocessing
- **Tools** Git/GitHub, TensorFlow Lite

Certifications

- **Python Data Visualization (Coursera)** *Rice University*
- **Machine Learning & Deep Learning (Coursera)** *IBM*
- **Python** *Certificate*
- **WordPress** *Certificate*
- **Open House FYP** *Certificate*

Additional Information

Coursework: Artificial Intelligence, Machine Learning, Deep Learning, NLP, Data Mining, Cloud Computing, Operating Systems, OOP, Web Engineering
Languages: English, Urdu
Interests: Research, Teaching, Medical AI, Smart Agriculture, Mobile App Development

References

1. **Dr. Uzair Iqbal**
   Tenured Associate Professor, Computer Science, CUI Islamabad Campus
   Email: uzair_iqbal@comsats.edu.pk
   Phone/Ext.: +92 51 90495351
2. **Dr. Muhammad Usman Yaseen**
   Assistant Professor, Computer Science, CUI Islamabad Campus
   Email: usman_yaseen@comsats.edu.pk
   Phone/Ext.: 5317